W03 L01

Distribution of Quadratic Forms

Recall,

Cochran's theorem

$\underset{\sim}{Y} \sim N(\underline{0}, I_n)$ and A is symmetric and idempotent.

Then $\underset{\sim}{Y}^T A \underset{\sim}{Y} \sim \chi^2_r$ where $r = \rho(A)$.

Result

$\underset{\sim}{Y} \sim N(\underline{0}, \Sigma)$, A is symmetric and $A\Sigma$ has eigenvalues 0 or 1.

Then $\underset{\sim}{Y}^T A \underset{\sim}{Y} \sim \chi^2_r$ where $r = \rho(A\Sigma)$.

Proof: $\Sigma = BB^T$ and $\underset{\sim}{Y} \overset{D}{=} B\underset{\sim}{Z}$ where $\underset{\sim}{Z} \sim N(\underline{0}, I)$.

$$\underset{\sim}{Y}^T A \underset{\sim}{Y} \overset{D}{=} \underset{\sim}{Z}^T B^T A B \underset{\sim}{Z}.$$

Note that the non-zero eigenvalues of $B^T AB$ is same as that of $ABB^T = A\Sigma$.

So, $B^T AB$ has the eigenvalue 1 with multiplicity $r = \rho(A\Sigma)$, and $(p - r)$ zero-eigenvalues.

This implies that the spectral decomposition of $B^T AB$ will be of the form:

$$B^T AB = UU^T \quad \text{with} \quad U^T U = I_r.$$

Thus,

$$\underset{\sim}{Y}^T A \underset{\sim}{Y} = \underset{\sim}{Z}^T B^T A B \underset{\sim}{Z} = \underset{\sim}{Z}^T U U^T \underset{\sim}{Z} = \|U^T \underset{\sim}{Z}\|^2.$$

$$\underset{\sim}{Z} \sim N(\underline{0}, I) \;\Rightarrow\; U^T \underset{\sim}{Z} \sim N(\underline{0}, I_r) \;\Rightarrow\; \|U^T \underset{\sim}{Z}\|^2 \sim \chi^2_r.$$

Result

$\underset{\sim}{Y} \sim N_p(\underset{\sim}{\mu}, \Sigma)$, $\Sigma > 0$. Then $(\underset{\sim}{Y} - \underset{\sim}{\mu})^T \Sigma^{-1} (\underset{\sim}{Y} - \underset{\sim}{\mu}) \sim \chi^2_p$.

Craig's theorem

$\underset{\sim}{Y} \sim N_p(\underline{0}, I)$ and A, B are two symmetric and idempotent matrices.

Then $\underset{\sim}{Y}^T A \underset{\sim}{Y} \perp\!\!\!\perp \underset{\sim}{Y}^T B \underset{\sim}{Y}$ if $AB = 0$.

Proof: $\underset{\sim}{Y}^T A \underset{\sim}{Y} = \|A\underset{\sim}{Y}\|^2$ and $\underset{\sim}{Y}^T B \underset{\sim}{Y} = \|B\underset{\sim}{Y}\|^2$.

Enough to show that $A\underset{\sim}{Y} \perp\!\!\!\perp B\underset{\sim}{Y} \;\Leftrightarrow\; AB = 0.$ [Property 6]

Example: $\underset{\sim}{X} \sim N_p(\underset{\sim}{\mu}, \sigma^2 I)$, $\sigma^2 > 0$.

$$S_y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2 \qquad (n-1)S_y^2 = \underset{\sim}{Y}^T\left(I - \frac{1}{n}\underline{1}\underline{1}^T\right)\underset{\sim}{Y}.$$

$$H_n = \left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right)$$

$$H_n\frac{1}{n} = \frac{1}{n} - \frac{1}{n}\mathbf{1}\mathbf{1}^T \cdot \frac{1}{n}\mathbf{1} = 0$$

Observe that $H_n^2 = H_n$.
Let $\underset{\sim}{Z} = \frac{1}{\sigma}(\underset{\sim}{Y} - \underset{\sim}{\mu}) \sim N(\underline{0}, I_n)$, then $\underset{\sim}{Z}^T H_n \underset{\sim}{Z} \sim \chi^2_{n-1}$.

$$\frac{1}{\sigma^2}(\underset{\sim}{Y} - \underset{\sim}{\mu})^T H_n (\underset{\sim}{Y} - \underset{\sim}{\mu}) \sim \chi^2_{n-1}.$$

Now,

$$(\underset{\sim}{Y} - \underset{\sim}{\mu})^T H_n (\underset{\sim}{Y} - \underset{\sim}{\mu}) = \underset{\sim}{Y}^T H_n \underset{\sim}{Y} - 2\underset{\sim}{\mu}^T H_n \underset{\sim}{Y} + \underset{\sim}{\mu}^T H_n \underset{\sim}{\mu}.$$

If $H_n\underset{\sim}{\mu} = 0$ then the above equation is same as $\underset{\sim}{Y}^T H_n \underset{\sim}{Y}$.

$$(\underset{\sim}{Y} - \underset{\sim}{\mu})^T H_n (\underset{\sim}{Y} - \underset{\sim}{\mu}) = \underset{\sim}{Y}^T H_n \underset{\sim}{Y}$$

If $\underset{\sim}{\mu} = c\,\underline{1}$ then

$$\frac{(n-1)S_Y^2}{\sigma^2} \sim \chi^2_{n-1}.$$

$$H_n = \left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right)$$

$$H_n\underline{1} = \underline{1} - \frac{1}{n}\underline{1}\underline{1}^T\underline{1} = 0.$$

Linear Regression

Recall the linear regression model with response y, and regressors/covariates $\{x_1, \dots, x_p\}$ and n obs. as follows:

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

where

$$\underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \underset{\sim}{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

We assume $E(\underset{\sim}{\varepsilon}) = 0$ and $\text{var}(\underset{\sim}{\varepsilon}) = \sigma^2 I$. Also, $\varepsilon_1, \dots, \varepsilon_n$ are independent.

Parameters: $(\underset{\sim}{\beta}, \sigma^2)$.

- $\underset{\sim}{\beta}$: vector of regression coefficients.
- σ^2: error variance.

$$\begin{cases} Y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \varepsilon_i, \quad i = 1, \dots, n & \text{estimate} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \\ Y_i = \gamma_0 + \gamma_1 x_{1i} + \varepsilon_{1i} & \\ Y_i = \gamma_0 + \gamma_2 x_{2i} + \varepsilon_{2i} & \\ \vdots & \text{estimate} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_p \end{pmatrix} \\ Y_i = \gamma_0 + \gamma_p x_{pi} + \dots + \gamma_p x_{pi} + \varepsilon_{pi} & \end{cases}$$

The above 2 estimates are not the same, they are modelling two different models hence the estimates won't be same.

Least Squares Method

We minimize

$$\underset{\sim}{\varepsilon}^T \underset{\sim}{\varepsilon} = (\underset{\sim}{Y} - X\underset{\sim}{\beta})^T (\underset{\sim}{Y} - X\underset{\sim}{\beta}) = \sum_{i=1}^{n} (Y_i - \beta_0 - \dots - \beta_p x_{pi})^2 = S(\underset{\sim}{\beta}),$$

w.r.t. $\underset{\sim}{\beta}$.

To minimize $S(\underset{\sim}{\beta})$ we differentiate $S(\underset{\sim}{\beta})$ w.r.t. $\underset{\sim}{\beta}$.

$$S(\beta) : \mathbb{R}^{p+1} \to \mathbb{R}$$

$$\frac{\partial S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}} = \frac{\partial}{\partial \underset{\sim}{\beta}}\Big(\|\underset{\sim}{Y}\|^2 + \underset{\sim}{\beta}^T X^T X \underset{\sim}{\beta} - 2\underset{\sim}{Y}^T X \underset{\sim}{\beta}\Big) = 0 + 2X^T X \underset{\sim}{\beta} - 2X^T \underset{\sim}{Y}.$$

First order condition:

$$\frac{\partial S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}} = 0 \;\Rightarrow\; X^T X \underset{\sim}{\beta} = X^T \underset{\sim}{Y}.$$

$$\underset{\sim}{\hat{\beta}} = (X^T X)^{-1} X^T \underset{\sim}{Y} \quad \text{when } (X^T X)^{-1} \text{ exists.}$$

Second order condition:

$$\frac{\partial^2 S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}\, \partial \underset{\sim}{\beta}^T} = X^T X.$$

When $(X^T X)^{-1}$ exists, then $X^T X > 0$.

$\Rightarrow$ $\underset{\sim}{\hat{\beta}}$ is the unique minima.

Useful derivatives, HW:

1. $\dfrac{\partial}{\partial x}(x^T A x)$ when A is symmetric.

$$= \frac{\partial}{\partial x_j}\Big(\sum_{i,k} a_{ik} x_i x_k\Big)_{j=1,\dots,p} = (A + A^T)x = 2Ax.$$

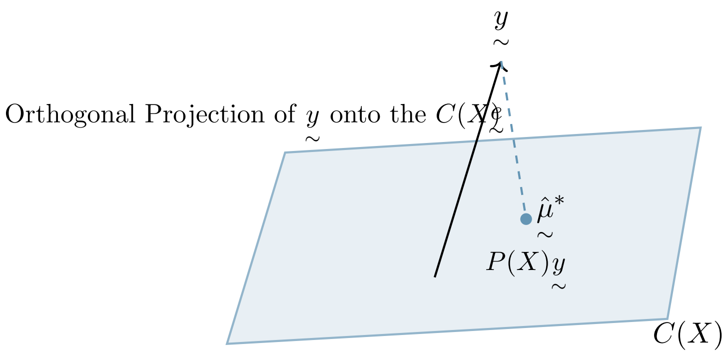
2. $\dfrac{\partial a^T x}{\partial x} = a.$

3. $\dfrac{\partial Ax}{\partial x} = A$ when A is symmetric.

Geometric interpretation

The problem is to find $\underset{\sim}{\hat{\beta}}$ s.t. $\|\underset{\sim}{y} - X\underset{\sim}{\hat{\beta}}\|^2$ is minimum.

i.e., to find the component $\underset{\sim}{\hat{\mu}}^*$ in the column space of X, $C(X)$, which is closest to $\underset{\sim}{y}$.



Orthogonal projection of $\underset{\sim}{y}$ onto $C(X)$; $C(X)$ is the plane.

$$a_0 \underset{\sim}{x}_{(0)} + a_1 \underset{\sim}{x}_{(1)} + \cdots + a_p \underset{\sim}{x}_{(p)} = X \underset{\sim}{a}$$

$$\underset{\sim}{\hat{\mu}}^* = P(X)\underset{\sim}{y} \quad \text{where } P(X) \text{ is the orthogonal projection matrix onto the } C(X).$$

Properties of Orthogonal projection matrix

- $P^2(X) = P(X) \;\Leftrightarrow\; P(X)\big[I - P(X)\big] = 0$ and also symmetric.
- If $\underset{\sim}{\mu} \in C(X)$ then $P(X)\underset{\sim}{\mu} = \underset{\sim}{\mu}$.
- $\rho\big(P(X)\big) = \rho(X) = \operatorname{tr}\big(P(X)\big)$.
- If $(X'X)^{-1}$ exists then $P(X) = X(X'X)^{-1}X'$.

  If $(X'X)$ is singular then $P(X) = X(X'X)^{-}X'$, where A^{-} is any g-inverse of A.

Least Square solution (Geometrically)

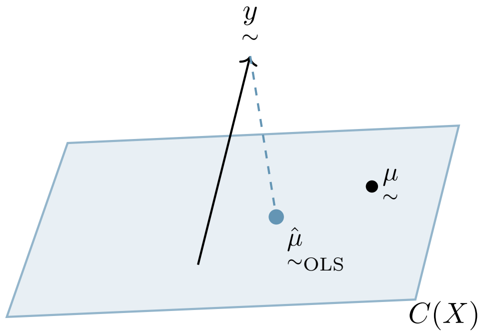
Let

$$\underset{\sim}{\hat{\mu}}_{\text{OLS}} = X(X'X)^{-1}X'\underset{\sim}{y} \in C(X) = P(X)\underset{\sim}{y}.$$

We will show that $\underset{\sim}{\hat{\mu}}_{\text{OLS}}$ is the closest point of $\underset{\sim}{y}$ in $C(X)$.

Take any $\underset{\sim}{\mu} \in C(X)$

$$\begin{aligned}
\|\underset{\sim}{y} - \underset{\sim}{\mu}\|^2 &= \|\underset{\sim}{y} - \underset{\sim}{\hat{\mu}}_{\text{OLS}} + \underset{\sim}{\hat{\mu}}_{\text{OLS}} - \underset{\sim}{\mu}\|^2 \\
&= \|\underset{\sim}{y} - \underset{\sim}{\hat{\mu}}_{\text{OLS}}\|^2 + \|\underset{\sim}{\hat{\mu}}_{\text{OLS}} - \underset{\sim}{\mu}\|^2.
\end{aligned}$$



$\hat{\underset{\sim}{\mu}}_{OLS}$ is the closest point of $\underset{\sim}{y}$ in $C(X)$.

$$
\begin{aligned}
\big(\underset{\sim}{y} - \hat{\underset{\sim}{\mu}}_{OLS}\big)^T\big(\hat{\underset{\sim}{\mu}}_{OLS} - \underset{\sim}{\mu}\big) &= \big(\underset{\sim}{y} - P(X)\underset{\sim}{y}\big)^T\big(P(X)\underset{\sim}{y} - \underset{\sim}{\mu}\big) \\
&= \underset{\sim}{y}^T\big(I - P(X)\big)\big(P(X)\underset{\sim}{y} - \underset{\sim}{\mu}\big) \\
&= \underset{\sim}{y}^T\underbrace{\big(I - P(X)\big)P(X)}_{0}\underset{\sim}{y} - \underset{\sim}{y}^T\underbrace{\big(I - P(X)\big)}_{\underset{\sim}{\mu} - P(X)\underset{\sim}{\mu}}\underset{\sim}{\mu} \\
&= 0 \quad \text{on } \underset{\sim}{\mu} \in C(X).
\end{aligned}
$$

Thus, the distance is minimized if $\hat{\underset{\sim}{\mu}}_{OLS} = \underset{\sim}{\mu}$.

So, the optimum point in $C(X)$ is:

$$\hat{\underset{\sim}{\mu}}_{OLS} = P(X)\underset{\sim}{y}.$$

Q. For which $\hat{\underset{\sim}{\beta}}$, $P(X)\underset{\sim}{y} = X\hat{\underset{\sim}{\beta}}$?